

Conley–Zehnder Indices and Bifurcations of the Spatial Rotating Kepler Problem

Dongho Lee

19 April, 2026

Seoul National University (SNU), Center for Quantum Structures in Modules and Spaces (QSMS)

Presented in Dalian University



This talk is based on my thesis and paper [Lee26], which is an extended result of [AFFvK13] dealt with the planar rotating Kepler problem.

- Basic properties of the rotating Kepler problem.
- **Result 1.** Description of the moduli space of periodic Kepler orbits using angular momentum and Laplace–Runge–Lenz vector.
- **Result 2.** Computation of Conley–Zehnder indices of periodic Kepler orbits.
- Further discussion about generic Hamiltonian bifurcations.

Rotating Kepler Problem :

Basics

Motivation: Circular Restricted Three-body Problem

The **circular restricted three-body problem** (CRTBP) describes a motion of a mass-less object with respect to a gravitational force of two bodies, which are in a motion of circular rotation

$$M(t) = (1 - \mu)(\cos t, -\sin t, 0), \quad E(t) = -\mu(\cos t, -\sin t, 0).$$

Using a rotating frame, we can write the Hamiltonian on $T^*\mathbb{R}^3 \setminus \{M, E\}$,

$$H(q, p) = \frac{1}{2}|p|^2 - \frac{\mu}{|q - M|} - \frac{1 - \mu}{|q - E|} + (q_1 p_2 - q_2 p_1).$$

where $M = (1 - \mu, 0, 0)$, $E = (-\mu, 0, 0)$.

Note. The rotating frame is expressed by adding the angular momentum to the Hamiltonian, $L_3 = q_1 p_2 - q_2 p_1$.

Circular Restricted Three-body Problem

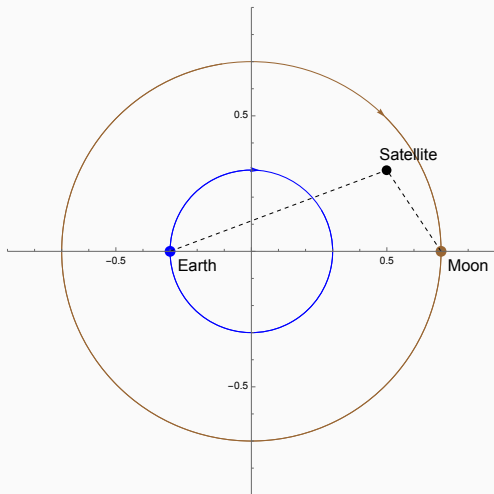


Figure 1: Simple illustration of CRTBP.

Rotating Kepler Problem

Rotating Kepler problem (RKP) is a limit case $\mu = 0$ of CRTBP defined by a Hamiltonian on $T^*\mathbb{R}^3 \setminus \{0\}$,

$$H = E + L_3 = \frac{1}{2}|p|^2 - \frac{1}{|q|} + (q_1p_2 - q_2p_1).$$

H : **Jacobi energy** (usually, $H = c$)

E : **Kepler energy**.

This describes a motion of a mass-less object under a gravitational force from the origin in a rotating frame. (e.g. a satellite near the Earth.)

Why is RKP important?

- Solution of CRTBP is impossible to compute analytically.
- Since RKP is *completely integrable*, we have complete solutions.
- Regarding CRTBP as a perturbation of RKP,

$$H_{CRTBP} = H_{RKP} + \mu G,$$

we can find out some features of CRTBP.

- For example, periodic orbits of RKP works as a backbone of CRTBP: the *retrograde*, *direct* and *vertical collision* orbits.

Hill's Region

The Hamiltonian of RKP can be written as

$$H = \frac{1}{2}|\tilde{p}|^2 - \left(\frac{q_1^2 + q_2^2}{2} + \frac{1}{|q|} \right) = \frac{1}{2}|\tilde{p}|^2 + U(q),$$

where $\tilde{p} = (p_1 - q_2, p_2 + q_1, p_3)$. We call $U(q)$ an **effective potential**.

Note that if $H(q, p) = c$, then $U(q) \leq c$.

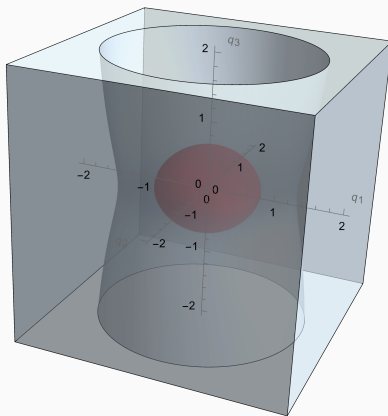
Hill's region : $\mathfrak{R}_c = \{q \in \mathbb{R}^3 : U(q) \leq c\} = \{q \in \mathbb{R}^3 : H(q, p) = c\}$.

The orbit with Jacobi energy c always remains in $\mathfrak{R}_c$.

- $\mathfrak{R}_c$ has 2 components (1 bounded, 1 unbounded) if $c < -3/2$.
- $\mathfrak{R}_c$ has 1 unbounded component if $c \geq -3/2$.

We call $c = -3/2$ a **critical energy**, and focus on the bounded component of $\mathfrak{R}_c$ for $c < -3/2$.

Hill's Region



The bounded component of $\mathfrak{R}_c$ for $c < -3/2$ is of contact type, and moreover the dynamics is a Finsler geodesic flow, proven in [CFvK14].

Kepler's Law

Let $E = |p|^2/2 - 1/|q| < 0$ be the Kepler Hamiltonian. We call the Hamiltonian orbits of E a **Kepler orbit**.

In 17th century, Kepler established these three laws.

1. Kepler orbits are ellipses with one focus at the origin.
2. The angular momentum $L = q \times p$ is a conserved quantity.
3. The period τ is determined by the Kepler energy,

$$\tau^2 = \frac{(2\pi)^2}{(-2E)^3}.$$

We will first discuss about conserved quantities of Kepler orbits, and then go to the rotating Kepler problem.

Conserved Quantities

Consider the Hamiltonian equation

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}.$$

We interpret this as a **Hamiltonian vector field**

$$X_H = \sum_i \frac{\partial H}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial H}{\partial q_i} \frac{\partial}{\partial p_i}.$$

A function F is a **conserved quantity** if $X_H(F) = 0$, i.e.,

$$\{F, H\} := \frac{\partial F}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial F}{\partial p_i} \frac{\partial H}{\partial q_i} = \frac{\partial F}{\partial q_i} \dot{q}_i + \frac{\partial F}{\partial p_i} \dot{p}_i = 0.$$

We call $\{F, H\}$ a **Poisson bracket**, which measures the conservation.

Conserved Quantities

Kepler problem has 6 conserved quantities.

Angular momentum $L = q \times p$.

- Direction of L = Normal to the plane which the orbit contained in.

Laplace–Runge–Lenz vector $A = p \times L - \frac{q}{|q|}$

- Direction of A = Direction of the major axis

Same components of L and A Poisson commutes, i.e. $\{L_i, A_i\} = 0$.

The $2n$ -dimensional Hamiltonian system with n -Poisson commuting conserved quantities is called **completely integrable**.

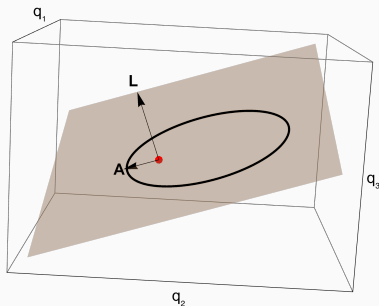
A completely integrable system has an **action-angle coordinates** on which the system is a linear flow on a torus, so can be solved analytically.

(cf. Arnold–Liouville theorem, [Arn89].)

Conserved Quantities

A Kepler orbit is determined by E , L and A , whose formula in the polar coordinate of $L \cdot q = 0$ is

$$r = \frac{|L|^2}{1 + |A| \cos(\theta - g)} \quad (g \text{ is determined by the direction of } A).$$



Note that the eccentricity is given by $\varepsilon^2 = |A|^2$.

Conserved Quantities

Important relation : $|A|^2 = 2E|L|^2 + 1$.

Proof.

$$\begin{aligned}|A|^2 &= |p \times L|^2 - \frac{2}{|q|} q \cdot p \times L + \frac{|q|^2}{|q|^2} = |p|^2 |L|^2 - \frac{2}{|q|} (q \times p \cdot L) + 1 \\ &= |p|^2 |L|^2 - \frac{2}{|q|} |L|^2 + 1 = \left(|p|^2 - \frac{2}{|q|} \right) |L|^2 + 1 = 2E|L|^2 + 1.\end{aligned}$$

□

So the eccentricity is $\varepsilon^2 = |A|^2 = 2E|L|^2 + 1$.

Periodic Solutions of RKP

Since $\{E, L_3\} = 0$, we can decompose the Hamiltonian flow of RKP by

$$Fl_t^H = Fl_t^E \circ Fl_t^{L_3}.$$

Since $Fl_t^{L_3}$ is a rotation along q_3 -axis, the solution of RKP is simply a composition of Kepler cylinder and q_3 -rotation.

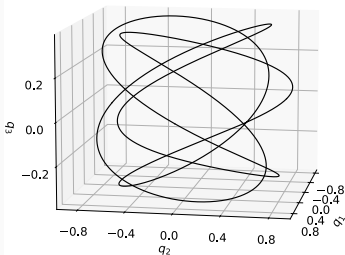
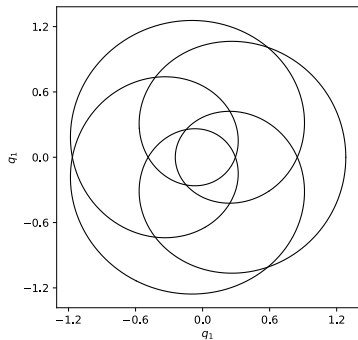
A resonance condition of two periods is required for to be periodic.

$$k\tau = 2\pi l \quad \Rightarrow \quad k \frac{2\pi}{(-2E)^{3/2}} = 2\pi l \quad \Rightarrow \quad E = E_{k,l} := -\frac{1}{2} \left(\frac{k}{l} \right)^{2/3}.$$

Thus if $E = E_{k,l}$, every Kepler orbit is periodic in RKP.

Of course, there are exceptions, which are more important.

Resonant Solutions



These are how planar and spatial resonant orbits look like.
(Figure in courtesy of Chankyu Joung.)

Retrograde and Direct Orbits

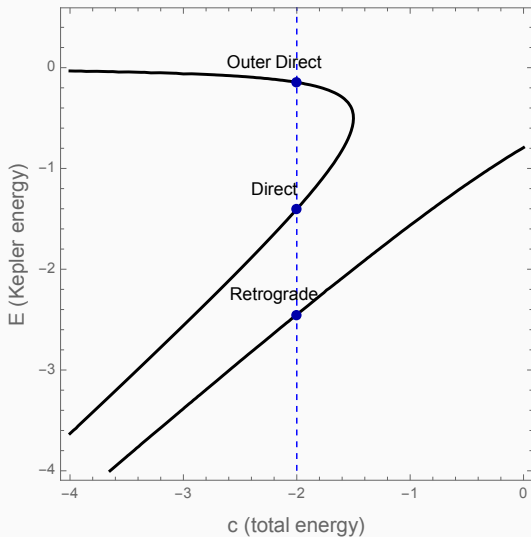
If a Kepler orbit is *planar* and *circular*, the composition with L_3 -flow cannot effect the periodicity of the orbit.

- Planar condition : $L = L_3$.
- Circular condition: $\varepsilon^2 = 2EL_3^2 + 1 = 2E(c - E)^2 + 1 = 0$.

For fixed $c < -3/2$, there are 3 **planar circular orbits** with different E .

1. **Retrograde orbit** γ_+ : $L_3 > 0$, smaller E and smaller radius.
2. **Direct orbit** γ_- : $L_3 < 0$, larger E and larger radius.
3. The rest one, outer direct orbit, lies on the unbounded component, and not of our interest.

Retrograde and Direct Orbits



Collision Orbits

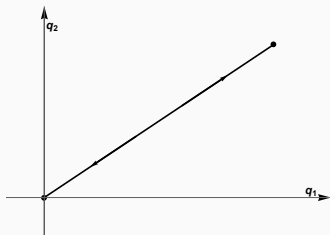
[Mos70] For $H < -3/2$, we can embed the Hamiltonian flow on the level set $H^{-1}(c)$ into a flow on T^*S^3 by **Moser regularization**.

For Kepler problem, it is embedded as a unit geodesic flow on S^3 .

[CFvK14] showed that RKP is embedded as a Finsler geodesic flow.

$\Rightarrow$ Compactification of the energy level set by $ST^*S^3 \simeq S^3 \times S^2$.

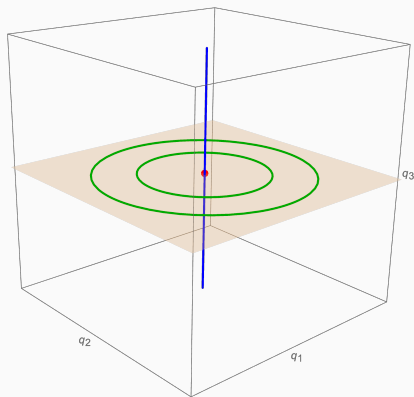
Collision orbits are added. ($\varepsilon = |A| = 1$, $L = 0$, $E = c$.)



This orbit bounds between two points; a shrunken ellipse.

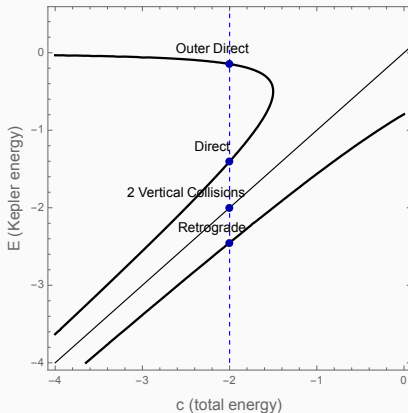
Vertical Collision Orbits

Vertical collision orbits $\gamma_{c\pm}$ are not effected by L_3 -flow, so it remains periodic in RKP.



Vertical Collision Orbits

Since $E = c$, we can mark the vertical collision orbits as follows.



Space of Periodic Orbits of RKP

Moduli Space of Kepler Orbits

Recall. E, L and A characterizes the Kepler orbit.

Denote $x = \sqrt{-2EL} - A$, $y = \sqrt{-2EL} + A$.

$$\Rightarrow |x|^2 = |y|^2 = -2E|L|^2 + |A|^2 = 1.$$

The moduli space of the Kepler orbits with Kepler energy E is

$$\mathcal{M}_E = \{(x, y) : |x|^2 = |y|^2 = 1\} \simeq S^2 \times S^2.$$

(Space of unit geodesics of S^3) = $ST^*S^3/S^1 \simeq S^2 \times S^2$.

Note. In the planar problem, the moduli space is $\mathbb{RP}^3/S^1 \simeq S^2$.

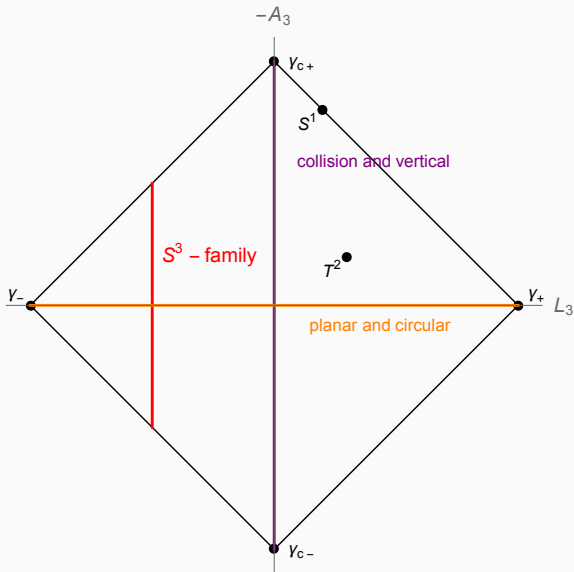
Properties of $\mathcal{M}_E$

$L_3 = (x_3 + y_3)/\sqrt{-2E}$ and $A_3 = (x_3 - y_3)/\sqrt{-2E}$ can be regarded as a function on $\mathcal{M}_E$.

1. $\gamma_{\pm}$ are minimum and maximum of L_3 .
2. $\gamma_{c\pm}$ are minimum and maximum of A_3 .
3. Every regular level set of L_3 and A_3 is S^3 . (Handle attachment)
4. $L_3^{-1}(0)$ is singular, homeomorphic to a suspension of S^2 .

$\Rightarrow$ Generally, $\Sigma_{k,l} = L_3^{-1}(c - E_{k,l})$ is topologically S^3 .

Properties of $\mathcal{M}_E$



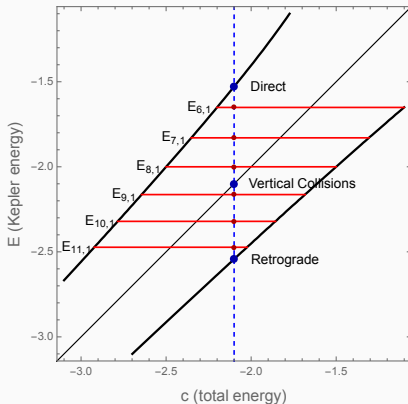
Bifurcation Structure

Let $c_{\pm}^{k,l} = E_{k,l} \pm 1/\sqrt{-2E_{k,l}}$. This is the Jacobi energy of retrograde/direct orbits with Kepler energy $E_{k,l}$.

1. At the energy level $c_{-}^{k,l}$, some cover of the direct orbit is contained in a $\Sigma_{k,l}$ -family.
2. Consider the resonance condition: $k\tau = 2\pi l$. The Kepler flow rotates k -times, while L_3 -flow rotates l -times. In total, we have $(k-l)$ -rotation, so we have $\gamma_{-}^{k-l} \in \Sigma_{k,l}$.
3. Similarly, at $c_{+}^{k,l}$ we have $\gamma_{+}^{k+l} \in \Sigma_{k,l}$.
4. In short, as varying c , an S^3 -family $\Sigma_{k,l}$ is born by γ_{-}^{k-l} at $c = c_{-}^{k,l}$, and dead by γ_{+}^{k+l} at $c = c_{+}^{k,l}$.
5. $\Sigma_{k,l}$ become singular at $c = E_{k,l}$, which consists of collision orbits.

Periodic Orbits in $H^{-1}(c)$

For generic energy level c , the energy hypersurface $H^{-1}(c)$ contains 4 nondegenerate orbits and (infinitely many) Morse–Bott S^3 -families.



To see bifurcation, we can see this graph in horizontal direction.

Conley–Zehnder Indices

Conley–Zehnder Index

The **Conley–Zehnder index** is, heuristically, a symplectic invariant of a periodic orbit which measures a winding number of a symplectic frame, which is given by an integer $\mu_{CZ}(\gamma) \in \mathbb{Z}$.

Let Ψ_t be the linearization Hamiltonian flow, which is a path of symplectic matrices.

$\mu_{CZ}(\gamma)$ is well-defined if $\det(\Psi_\tau - \text{Id}) \neq 0$ where τ is the period of γ . This condition is called **non-degeneracy** of γ .

1. Consider a change of the energy level $H = c$.
2. If γ_0 is non-degenerate and $H(\gamma) = c_0$, then there exists an *orbit cylinder* γ_c parametrized by c . CZ index is constant along γ_c .
3. When γ_c degenerates, the orbit cylinder breaks, and index changes.
4. This is called **bifurcation**: birth or death of periodic orbits.

What is Floer homology?

A **Hamiltonian Floer homology** is a homology defined on a symplectic manifold W by a chain complex $CF(W, H)$, where

- generators are 1-periodic orbits, graded by Conley–Zehnder indices.
- differential is defined by counting of *Floer cylinders*, which are solutions of a PDE with boundaries asymptotic to periodic orbits.

Fact. If W is closed, $HF_*(W, H) \simeq H_*(W)$. (independent of H !)

This can be regarded as a Morse homology on a free loop space with *symplectic action functional*. See [AD14] for details.

Variants of Floer Homology

1. Local Floer homology

We can compute $HF(H)$ on a small neighborhood of a specific periodic orbit γ , which does not contain any other orbits.

This is invariant under small perturbation of the Hamiltonian (cf.[GG10]).

$\Rightarrow$ A symplectic invariant which can track the bifurcation.

2. Symplectic homology

For open symplectic manifold, we define a **symplectic homology** $SH(W)$ as a direct limit of Hamiltonian Floer homology.

Periodic orbits in a specific energy hypersurface $H^{-1}(c)$ can be treated in this way, where $SH(W)$ consists of periodic orbits of *any* period.

Fact. [Vit99] $SH_*(T^*M; \mathbb{Q}) \simeq H_*(\mathcal{LM}, \mathbb{Q})$ with a degree shift.

Conley–Zehnder Index of Planar Circular Orbits

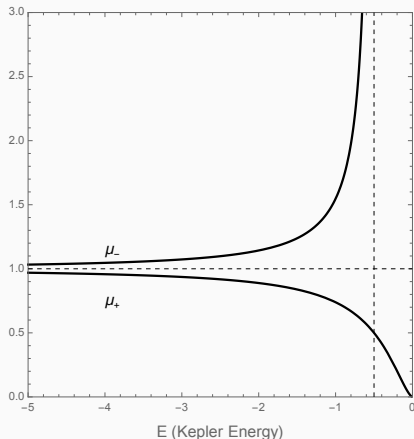
Theorem

Let $\gamma_{\pm}$ be the retrograde and direct orbits of Kepler energy E where $E \neq E_{k,l}$ for any k, l . Then $\gamma_{\pm}$ and their multiple covers are non-degenerate. The Conley–Zehnder index of N -th iterate of $\gamma_{\pm}$ is

$$\begin{aligned}\mu_{CZ}(\gamma_{\pm}^N) &= 2 + 4 \max \left\{ n \in \mathbb{Z}_{>0} : n < N \frac{(-2E)^{3/2}}{(-2E)^{3/2} \pm 1} \right\} \\ &= 2 + 4 \left\lfloor N \frac{(-2E)^{3/2}}{(-2E)^{3/2} \pm 1} \right\rfloor := 2 + 4 \lfloor N \mu_{\pm}(E) \rfloor.\end{aligned}$$

The index is exactly the twice compare to the planar problem, which was computed in [AFFvK13].

Conley-Zehnder Index of Planar Circular Orbits



The index of $\gamma_{\pm}^N$ is initially $4N \mp 2$, and changes by ∓ 4 whenever $\mu_{\pm}$ touches $k/N \Leftrightarrow E = E_{N \mp k, k}$. (Bifurcation occurs)

Conley-Zehnder Index of Vertical Collision Orbits

Theorem

Let $\gamma_{c\pm}$ be the vertical collision orbits of Kepler energy E where $E \neq E_{k,l}$ for any k, l . Then $\gamma_{c\pm}$ and their multiple covers are non-degenerate. The Conley-Zehnder index of N -th iteration of $\gamma_{c\pm}$ is

$$\mu_{CZ}(\gamma_{c\pm}^N) = 4N.$$

In particular, change of the energy does not change the index.

Interpretation by Symplectic Homology

$$SH_*^{+,S^1}(T^*S^3; \mathbb{Q}) \simeq \begin{cases} \mathbb{Q} & * = 2, \\ \mathbb{Q}^2 & * = 2k \geq 4, \\ 0 & \text{otherwise.} \end{cases}$$

For fixed N , there exists $c \ll -3/2$ such that $H^{-1}(c)$ consists of

1. $k(\leq N)$ -th covers of $\gamma_{\pm}$ of index $4k \mp 2$. (No bifucation)
2. Higher covers have index $> 4N + 2$.

Up to degree $4N + 2$, we have

1. One generator at degree 2. ($\gamma_{+ \cdot}$)
2. Two generators at degree 6, 10, 14, $\dots$, $4N + 2$. (γ_{+}^{k+1} and γ_{-}^k .)
3. Two generators at degree 4, 8, 12, $\dots$, $4N$. (γ_{c+}^k and γ_{c-}^k .)

This describes $SH_*^{+,S^1}(T^*S^3)$ up to degree $4N + 2$ completely.

Morse-Bott Property

$\Sigma_{k,l}$ -families: We use **Morse-Bott spectral sequence**.

$\Rightarrow$ We need Morse-Bott property (kind of non-degeneracy), and must compute the linearized return map.

We used two different action-angle coordinates:

1. Delaunay coordinate : $(p_l, p_g, p_\theta) = (1/\sqrt{-2E}, |L|, L_3)$.

Works for planar problem ([AFFvK13]), but degenerates at every planar orbit in the spatial case.

2. **LRL coordinate** : $(p_l, p_\eta, p_\theta) = (1/\sqrt{-2E}, A_3, L_3)$.

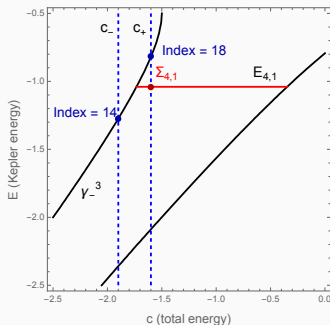
Also degenerates at some orbits, but covers planar orbits.

Index of Degenerate Orbits

Theorem

Index of S^3 -family $\Sigma_{k,l}$ with Kepler energy $E_{k,l}$ is

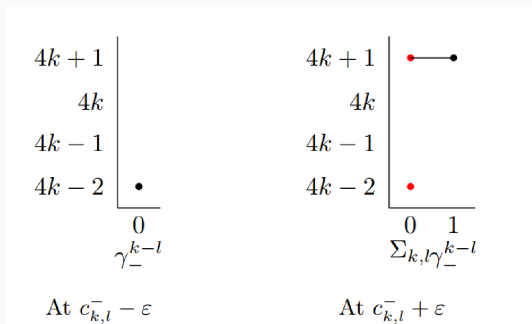
$$\begin{aligned}\mu_{RS}(\Sigma_{k,l}) &= \text{shift}(\Sigma) + \dim S^3/2 \\ &= (4k - 2) + 3/2 = 4k - 1/2.\end{aligned}$$



Index of Degenerate Orbits

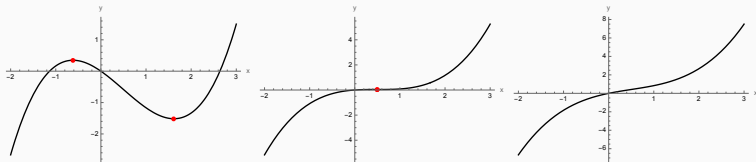
Previous results + local invariance of the symplectic homology.

We use *Morse–Bott spectral sequence* for the argument.

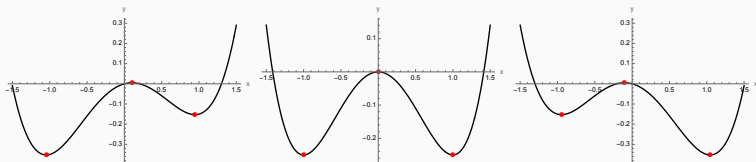


Bonus : Generic Hamiltonian Bifurcations

Generic Bifurcations of Morse Function



Birth-death bifurcation of critical points ($f_t(x) = x^3 - x^2 + tx$).



Handle-slide of critical points ($f_t(x) = x^4 - x^3 - x^2 + tx$).

Generic Bifurcations of Periodic Orbits

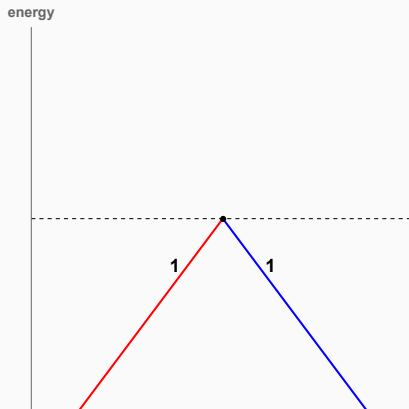
Meyer [Mey70] classified generic bifurcations of Hamiltonian periodic orbits into 5 types. (cf. [AM78] Ch.8)

1. Bifurcation occurs when the orbit cylinder breaks, i.e. there exists a eigenvalue of linearized return map such that $\lambda = 1$.
2. If $\lambda^k = 1$, k -th cover of the orbit bifurcates.
3. Find fixed points of an iterated return map. (Birkhoff normal form)

Note. The case of Hamiltonian orbit is different from the one of Morse function, because of multiple covers.

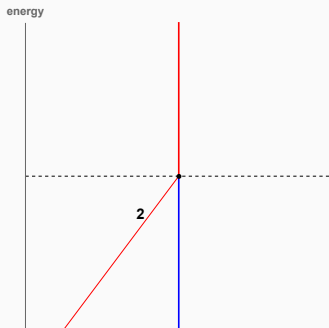
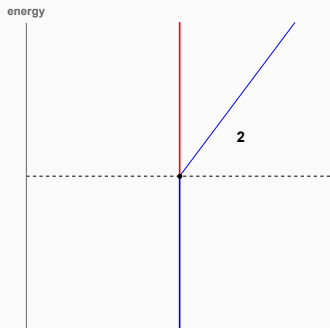
Birth-death

Birth-death: bifurcation of simple orbits.



Period Doubling and Materialization

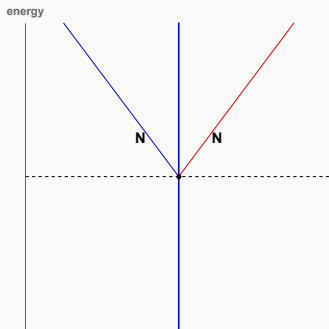
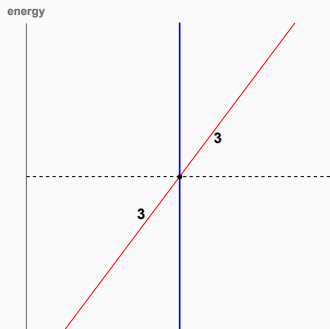
Period doubling / Materialization: bifurcation of double cover.



Phantom Kiss and Emission

Phantom kiss (or touch-and-go): bifurcation of triple or 4-th cover.

Emission : bifurcation of $N(\geq 4)$ -th cover.



If the system possesses a symmetry, there are other types.

Ex. CRTBP has $\mathbb{Z}_2$ -symmetry, which leads to a pitchfork bifurcation.

Local Floer Chain Complex

Recall. Degeneracy of an orbit = bifurcation = change of μ_{CZ} .

However, $HF_{loc}(W)$ is invariant through a bifurcation.

⇒ Bifurcation occurs a mutation of a chain complex.

- The number of generators changes.
- The degree (= CZ index) also changes.
- Differential map is re-arranged to make the homology consistent.

The simplest example: Two orbits of the birth-death must have index difference 1 to cancel each other.

Goal: Describe the explicit change of the chain complex through generic bifurcation types. (ongoing work with Hong-kwon Cho (SNU)).

Thank you for your attention!





Michèle Audin and Mihai Damian.

Morse theory and Floer homology.

Universitext. Springer, London; EDP Sciences, Les Ulis, 2014.

Translated from the 2010 French original by Reinie Ern .



Peter Albers, Joel W. Fish, Urs Frauenfelder, and Otto van Koert.

The Conley-Zehnder indices of the rotating Kepler problem.

Math. Proc. Cambridge Philos. Soc., 154(2):243–260, 2013.



Ralph Abraham and Jerrold E. Marsden.

Foundations of mechanics.

Benjamin/Cummings Publishing Co., Inc., Advanced Book Program,
Reading, MA, second edition, 1978.

With the assistance of Tudor Ra iu and Richard Cushman.



V. I. Arnold.

Mathematical methods of classical mechanics, volume 60 of Graduate Texts in Mathematics.

Springer-Verlag, New York, [1989?].

Translated from the 1974 Russian original by K. Vogtmann and A. Weinstein, Corrected reprint of the second (1989) edition.



Kai Cieliebak, Urs Frauenfelder, and Otto van Koert.

The Finsler geometry of the rotating Kepler problem.

Publ. Math. Debrecen, 84(3-4):333–350, 2014.



Viktor L. Ginzburg and Başak Z. Gurel.

Local Floer homology and the action gap.

J. Symplectic Geom., 8(3):323–357, 2010.



Dongho Lee.

Conley–zehnder indices of spatial rotating kepler problem.

Journal of Topology and Analysis, 0(0):1–40, 2026.



K. R. Meyer.

Generic bifurcation of periodic points.

Transactions of the American Mathematical Society, 149(1):95–107, 1970.



J. Moser.

Regularization of Kepler's problem and the averaging method on a manifold.

Comm. Pure Appl. Math., 23:609–636, 1970.



C. Viterbo.

Functors and computations in Floer homology with applications. I.

Geom. Funct. Anal., 9(5):985–1033, 1999.